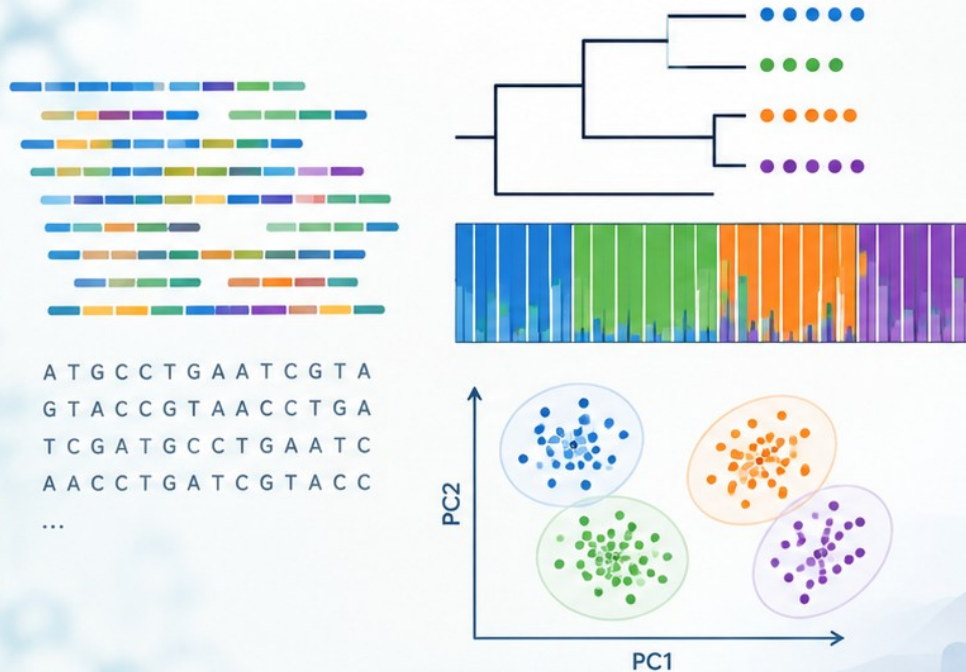


Genetic inference of population admixture

PR Stats Course

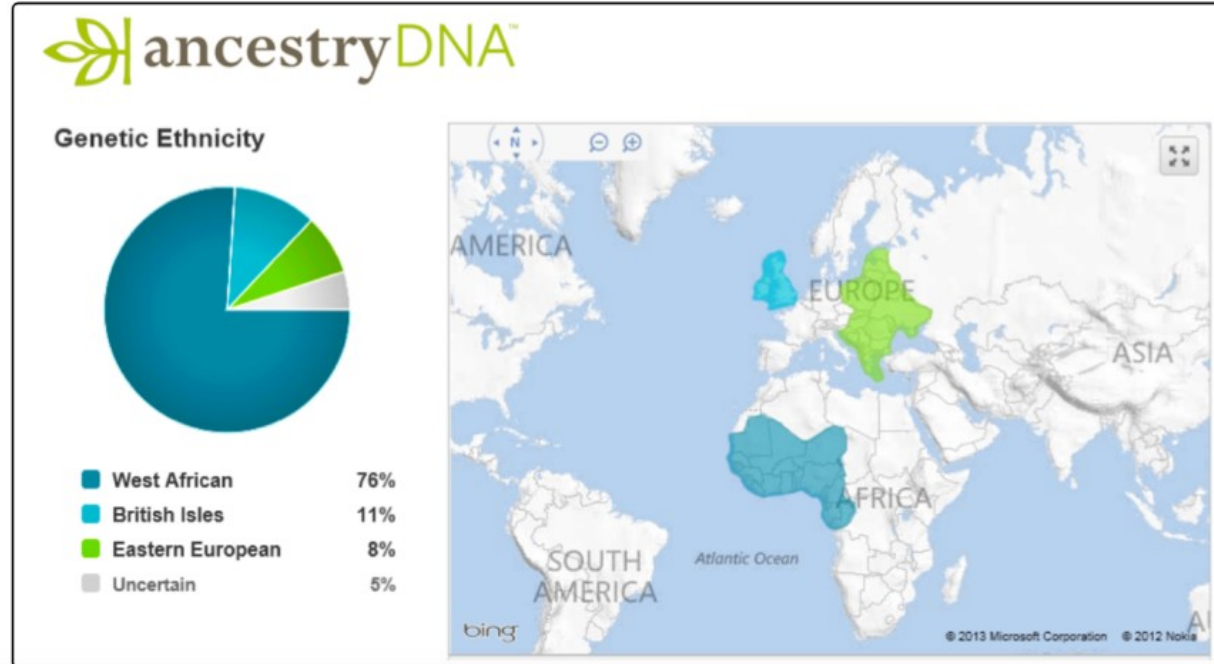
Nikolay Oskolkov, Group Leader (PI) at NIRI, Riga, Latvia



- ▶ Can we cluster samples into distinct groups based on their DNA
- ▶ Are specific samples admixed?
- ▶ If so what are the ancestral populations?
- ▶ And how much has been inherited from each of the ancestral populations?
- ▶ ... and related questions

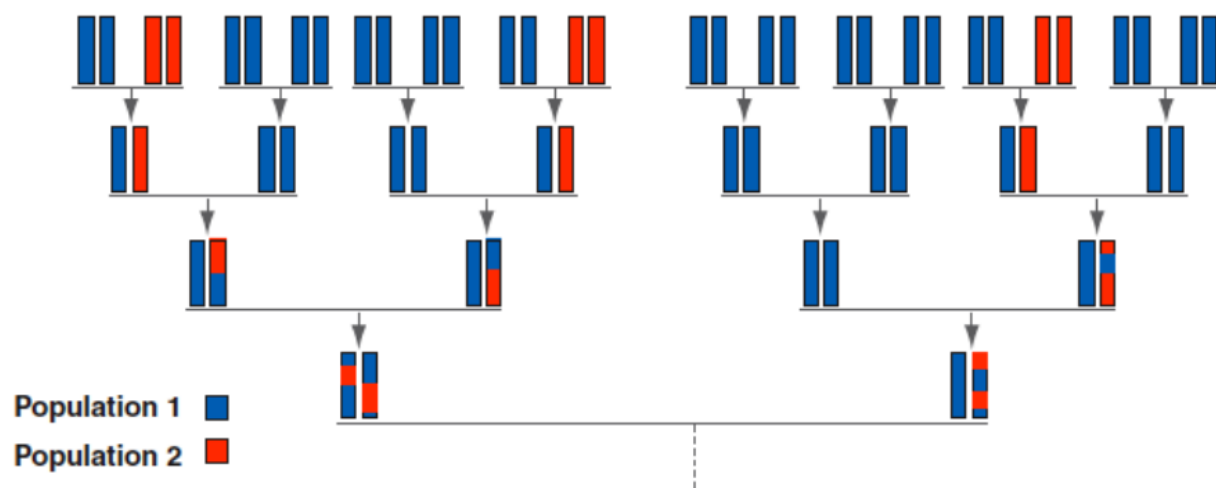


- ▶ Can be of interest in itself, e.g.
 - ▶ to learn about the demographic history of a population
 - ▶ to learn about the your own history



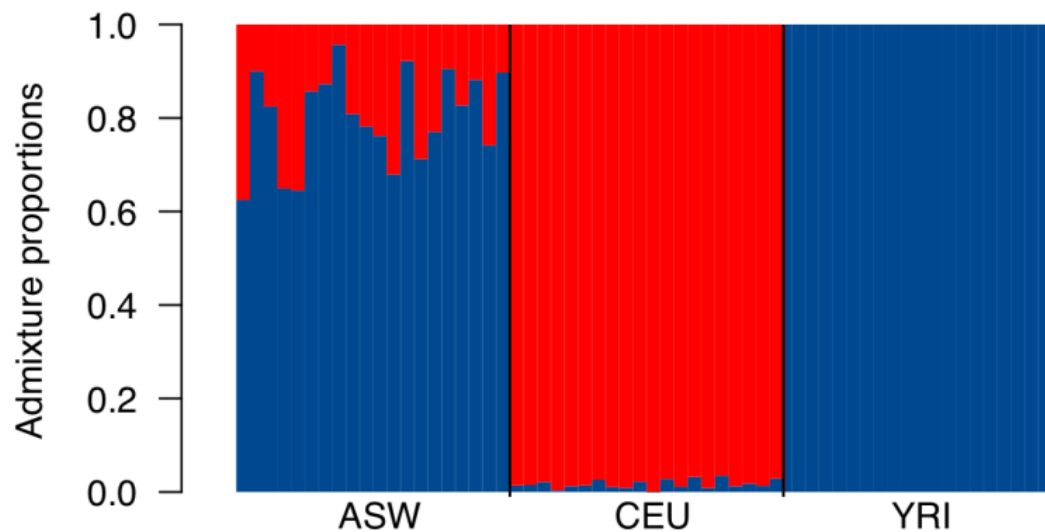
- ▶ But also used for QC and to ensure analyses are performed properly, e.g.
 - ▶ if not properly dealt with can lead to a lot of false positives in GWAS

- ▶ **Underlying assumption:** All analyzed individuals have DNA that originates from one or more of $K \geq 1$ ancestral source populations.
- ▶ **Consequence:** they will have some proportion of DNA from each of the K populations; these proportions are called admixture proportions
- ▶ **Examples with $K=2$:**
 - ▶ Samples from 2 unadmixed populations, e.g. European and Chinese
 - ▶ Two populations that have mixed, e.g. African Americans



- ▶ **Goal:** to infer admixture proportions (+ identify clusters)

- ▶ Several methods:
 - ▶ Maximum Likelihood (ML): e.g ADMIXTURE (Alexander et al 2009)
 - ▶ Bayesian: e.g. STRUCTURE (Pritchard et al. 2000)
- ▶ They all base their inference on called genotypes G and infer
 1. Admixture proportions, Q , e.g.:



2. Allele frequencies for all loci for all K populations, F

Q and F outputs of admixture analysis

G (genotypes):

Ind 1	A	T	G	T	T	A	A	T
	T	T	G	C	T	G	T	T
Ind 2	T	T	C	T	T	G	A	G
	T	G	C	T	A	G	T	T
...	T	G	G	T	T	G	A	G
	T	T	C	C	T	G	T	T
	A	T	G	T	T	A	A	T
	T	T	G	T	T	G	T	T
	T	T	G	T	A	G	A	G
	T	G	G	T	T	G	T	G
SNPs								

Q (admixture proportions):

Ind 1	4/16	12/16
Ind 2	6/16	10/16
...	3/16	13/16
	11/16	5/16
	3/16	13/16
	Pop1	Pop2

F (allele frequencies):

Pop1	2/5	3/4	4/4	3/3	3/3	0/3	2/3	0/2
Pop2	0/5	4/6	3/6	5/7	5/7	2/7	3/7	6/8
SNPs								

Likelihood function (1 individual i , 1 diallelic locus j)

Assume K source populations and let

- ▶ A and B denote the possible alleles at locus j
- ▶ G_{ij} be the genotype of i at locus j (measured in counts of allele A)
- ▶ $F^j = (f^{j1}, f^{j2}, \dots, f^{jK})$ denote the allele frequencies of allele A
- ▶ $Q^i = (q^{i1}, q^{i2}, \dots, q^{iK})$ denote i 's genome-wide admixture proportions

Then the likelihood function is $P(G_{ij}|Q^i, F^j)$. And to get that we notice

- ▶ for one of i 's allele copies: $P(A|Q^i, F^j) = q^{i1}f^{j1} + q^{i2}f^{j2} \dots + q^{iK}f^{jK} = h^{ij}$
- ▶ and therefore $P(B|Q^i, F^j) = 1 - h^{ij}$
- ▶ so assuming HWE we get the likelihood function:

$$P(G_{ij}|Q^i, F^j) = \begin{cases} (h^{ij})^2 & \text{if } G_{ij} = 2 \text{ (AA),} \\ 2h^{ij}(1 - h^{ij}) & \text{if } G_{ij} = 1 \text{ (AB),} \\ (1 - h^{ij})^2 & \text{if } G_{ij} = 0 \text{ (BB).} \end{cases}$$

- ▶ notice that if individual i e.g. has genotype AA:
then if the frequency of A is higher in a population, p , the likelihood increases the higher q^{ip} is, so we're capturing what we wanted

NGSadmix vs. ADMIXTURE

JOURNAL ARTICLE

Estimating Individual Admixture Proportions from Next Generation Sequencing Data

Line Skotte, Thorfinn Sand Korneliussen, Anders Albrechtsen Author Notes

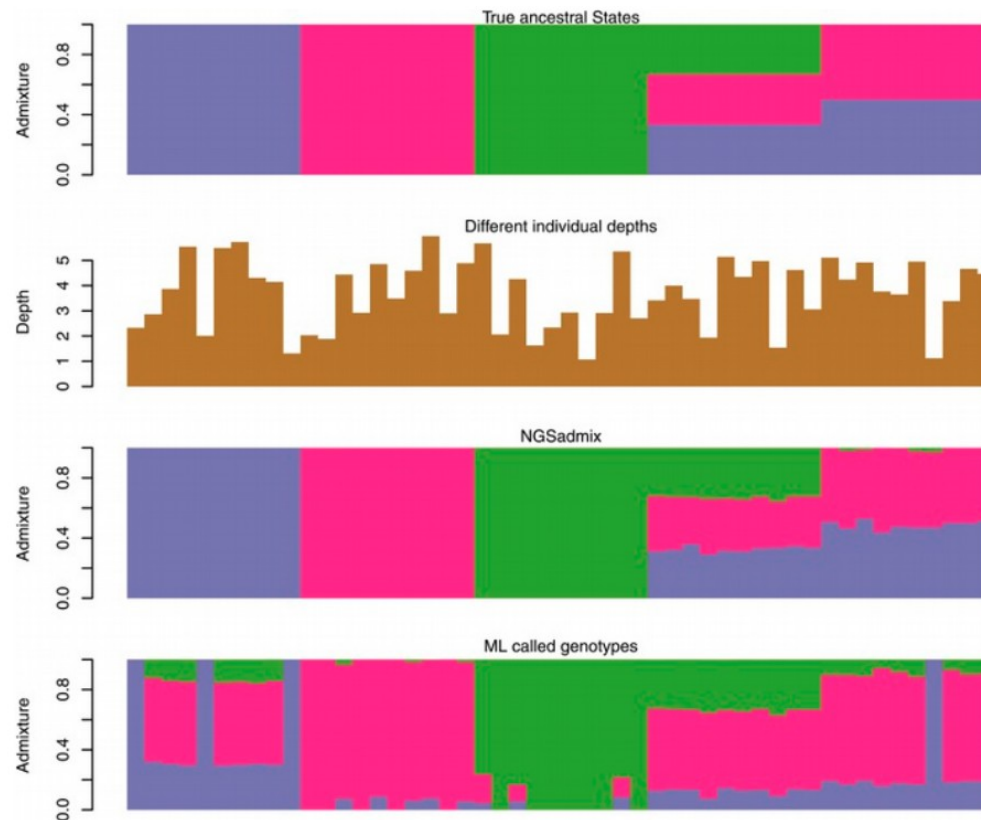
Genetics, Volume 195, Issue 3, 1 November 2013, Pages 693–702, <https://doi.org/10.1534/genetics.113.154138>

Published: 01 November 2013 Article history

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Abstract

Inference of population structure and individual ancestry is important both for population genetics and for association studies. With next generation sequencing technologies it is possible to obtain genetic data for all accessible genetic variations in the genome. Existing methods for admixture analysis rely on known genotypes. However, individual genotypes cannot be inferred from low-depth sequencing data without introducing errors. This article presents a new method for inferring an individual's ancestry that takes the uncertainty introduced in next generation sequencing data into account. This is achieved by working directly with genotype likelihoods that contain all relevant information of the unobserved genotypes. Using simulations as well as publicly available sequencing data, we demonstrate that the presented method has great accuracy even for very low-depth data. At the same time, we



Example of admixture plot from 1000G project

We will use ANGSD (Korneliussen T, 2014) and compute genotype likelihoods on 435 bam-files from the 1000G project.

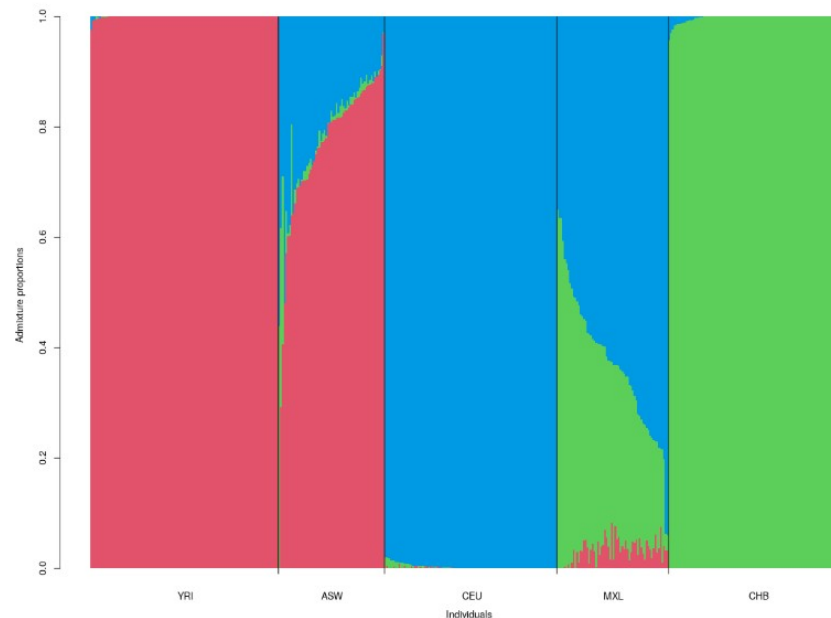
```
1 cd /home/nikolay/Documents/Teaching/PopGen_2023/data/1000G_bam_files
2 find $PWD -name '*.bam' > 1000G_bam_list.txt
3 angsd -bam 1000G_bam_list.txt -GL 2 -doMajorMinor 1 -doMaf 1 -SNP_pval 2e-6 \
4 -minMapQ 30 -minQ 20 -minInd 25 -minMaf 0.05 -doGLf 2 -out 1000G -P 5
```

Next, we will run NGSadmix to infer admixtures proportions:

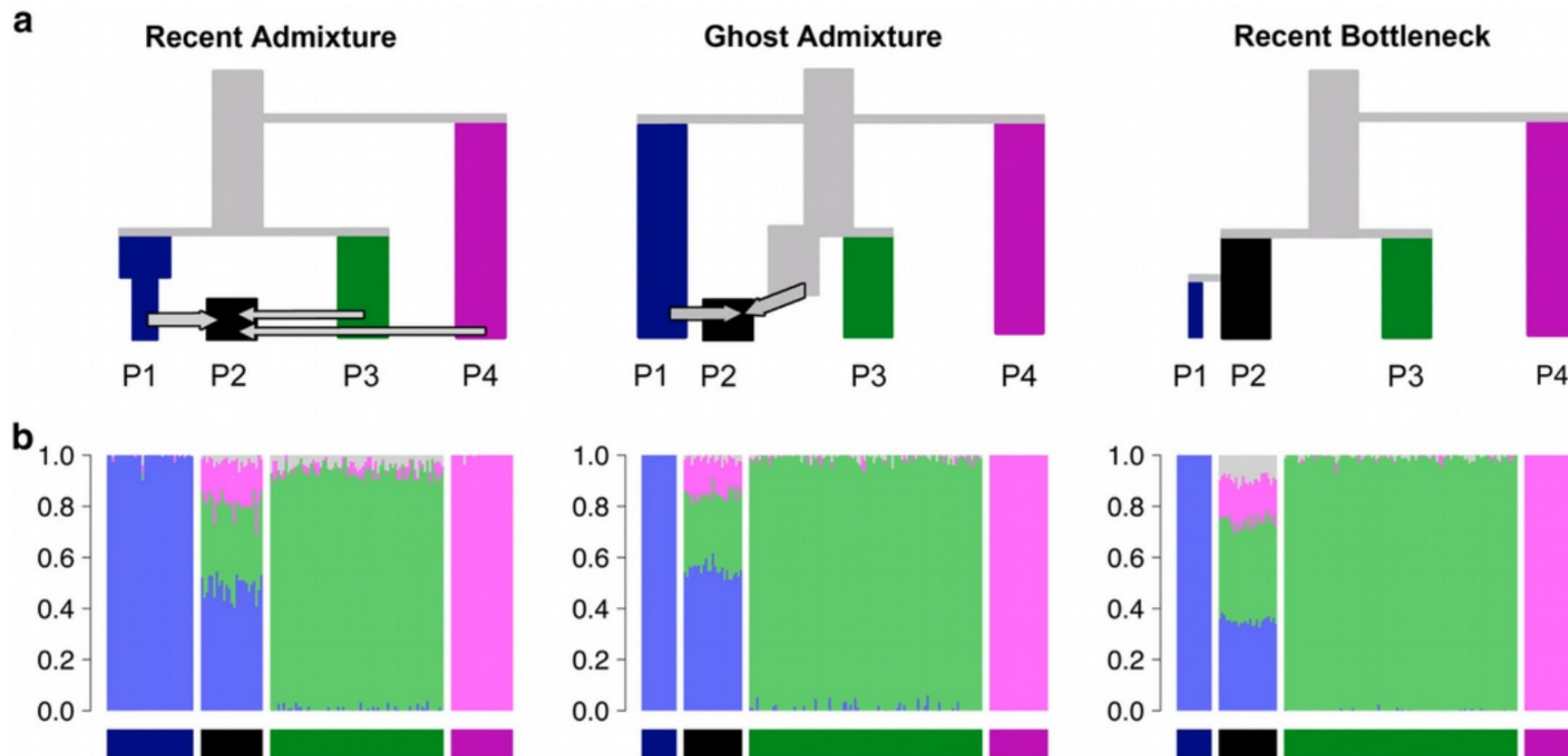
```
1 NGSadmix -likes 1000G.beagle.gz -K 3 -minMaf 0.05 -seed 1 -o 1000G
```

and finally we will plot the results using custom R scripts:

```
1 pops <- readLines("1000G_bam_list.txt")
2 pops <- sapply(strsplit(pops, "\\."), function(x) x[6])
3 source("https://raw.githubusercontent.com/GenisGE/evalAdmix/master/visFuns.R")
4 qopts <- read.table("1000G.qopt")
5 ord <- orderInds(pop = pops, q = qopts, popord = c("YRI", "ASW", "CEU", "MXL",
6 "CHB"))
7 barplot(t(qopts)[, ord], col = c(3, 2, 4), las = 2, space = 0, border = NA)
8 text(sort(tapply(1:length(pops), pops[ord], mean)), -0.05, unique(pops[ord]))
9 abline(v = cumsum(sapply(unique(pops[ord]), function(x) {
10   sum(pops[ord] == x)
11 })), col = 1, lwd = 1.2)
```



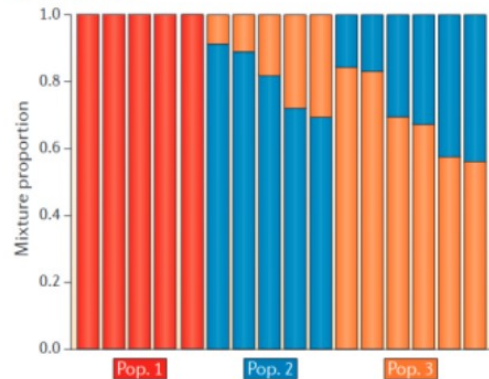
Admixture bar plots should not be over-interpreted



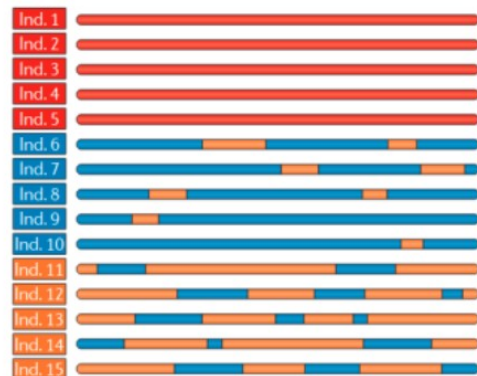
(Lawson DJ, 2018)

Haplotype clustering

e STRUCTURE



f Chromosome painting



Clustering
fineSTRUCTURE



Painting Profile

Population structure

- Ancestry assignments in STRUCTURE / ADMIXTURE do not identify where admixture has occurred
- Haplotype-based methods explore **local ancestry**, LD, recombination and subsequently fine-scale patterns of population structure at different scales
- Chromosome painting: the genome is a mosaic of LD blocks separated by recombination
- ChromoPainter and fineSTRUCTURE are tools for resolving **subtle differences** between populations.

(D. Lawson, 2012)

(Schraiber & Akey, 2015)

